

Sivan Syed

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Austin, TX

🌐 <https://ilgneous.github.io/>

Education

The University of Texas at Austin (GPA: 3.8/4.0)

May 2027

B.S. Electrical Engineering, B.S. Physics, B.S. Mathematics, Certificate German

Relevant Coursework: Stochastic Processes, Machine Learning, Kinematics & Dynamics, Embedded Systems

Graduate Coursework: Computational Physics, Magnetic Materials and Devices, Quantum Mechanics I

Experience

Emerson, Electrical Engineering Design Intern

May 2026–Aug 2026

- Modeled a 4–20 mA op-amp and potentiometer signal-conditioning schematic enabling field calibration of set-points of a reverse-engineered raw production PCB

Adom Industries Inc., R&D Design Engineer

May 2025–Aug 2025

- Architected a 48V VCU (RP5+RP2040) for ROS path planning via CAN
- Designed custom PCBs for LiDAR and motor control, applying digital isolators to suppress 48V EMI
- Applied iterative FEA to analyze stress, strain, and deformation, reducing material usage 88

Sandia National Laboratories, R&D Computational Engineer

May 2024–Aug 2024

- Built a fast current-pulse generator modeling capacitor discharges, synthesizing 10k+ waveforms for ML surrogate modeling and Uncertainty Quantification
- Trained an FNN surrogate model replicating magneto-hydrodynamics trajectory simulations at >98% fidelity
- Analyzed fluid RTIs and RMIs to diagnose simulation inaccuracies
- Wrote ODE trajectory solvers to track radial liner implosions and compare observed instability spectra

Banerjee Lab – 2D Material Synthesis & Characterization, Undergraduate Researcher

Sep 2025–Present

- Characterized and grew MoS₂ monolayers via CVD to optimize carrier type, doping, and bandgap
- Employed XRD, Raman, and Photoluminescence spectroscopy for material analysis

FSAE Longhorn Racing, Dynamics + Vehicle Modeling Engineer, Steering Lead

Sep 2022–Aug 2025

- Modeled vehicle dynamics in a Model-in-the-Loop simulation, validating slip-angle targets against track telemetry
- Designed a compliance model simulating part stress with FEM collision detection

Projects

SOT-Driven Skyrmion Racetrack Memory Device

- Designed a novel SOT-driven magnetic skyrmion racetrack memory in MuMax3, using a passive defect channel and VCMA gate to pin, release, and steer skyrmions on demand
- Automated a 33-job micromagnetic sweep on TACC Lonestar6 (A100 GPUs) via SLURM, diagnosing device failure mode through topological-charge and velocity analysis

Edge-ML UAV Flight Controller

- Designed a 10-Dof Quadcopter Dynamics simulator to optimize PID gains
- Architecting a deterministic 400 Hz attitude-control loop and a 50 Hz on-board anomaly-detection task under RTIC on an ARM Cortex-M4, isolating safety-critical control from inference
- Designing a cascaded PID controller on a hover-linearized rigid-body model with a nonlinear Kalman filter fusing noisy IMU measurements, prioritizing stability margins against latency and unmodeled dynamics

Embedded Racing Simulator

- Implemented multi-node CAN bus network, debugging bus arbitration and extending the data payload to support higher-bandwidth real-time telemetry between the steering wheel, pedal box, and central compute

Time-Series Damage Forecasting

- Forecast damage with an LSTM at >98% accuracy for optimal insurance pricing, benchmarking ARIMA, LSTM, and Transformer models head-to-head

Skills

Languages: C, C++, ARM, Rust, Python, MATLAB, Bash, JS, HTML, CSS, LaTeX

Tools: KiCAD, Cadence, FLASH, Draftsight, LTSpice, MuMax3, Fusion360, SOLIDWORKS, Ansys